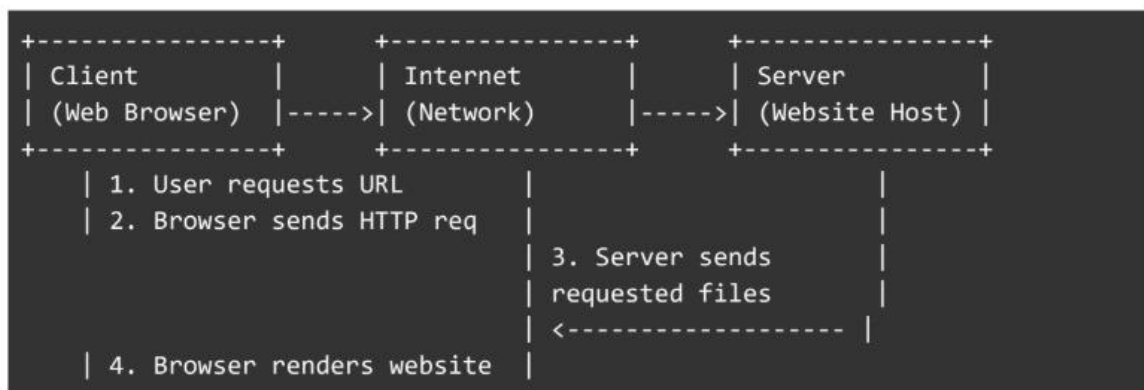


Assignment

Q1.

Frontend is the visible part of a website that users see and interact with, like buttons, images, and menus, made using HTML, CSS, and JavaScript. Backend is the hidden part that manages data, logins, and processes using tools like Python or Node.js. A full-stack developer handles both frontend and backend, building complete websites using stacks like MERN.

Q2.



The above diagram showing how the client-server model works in web architecture.

When a user sends a request (like searching something on Google), the browser (client) communicates with the server. The server processes the request, fetches the required data, and sends it back to the browser, which displays the results.

Q3.

When a user enters a URL, the browser sends an HTTP request to the web server hosting that site. The server finds and returns the webpage's HTML file. The browser then downloads related CSS,

JavaScript, and image files. The rendering engine constructs the DOM (Document Object Model), applies styles, executes scripts, and finally displays the fully rendered webpage to the user.

Q4.

Tools for Web Development and Their Uses

1. VS Code:

A simple and powerful code editor used to write HTML, CSS, and JavaScript. It supports many extensions to make coding easier.

2. Node.js & npm:

Node.js lets you run JavaScript outside the browser.

npm (Node Package Manager) helps you download and manage extra tools or libraries for your project.

3. Git & GitHub:

Git helps you save different versions of your code.

GitHub is an online platform where you can store your code and work together with others.

4. Browser Developer Tools:

These are built into browsers like Chrome or Firefox.

They help you check, test, and fix problems in your web pages.

5. Local Server:

These tools let you run and test your backend code on your own computer before uploading it online.

6. Postman:

Used to test and check APIs (how your website talks to a server).

Q5.

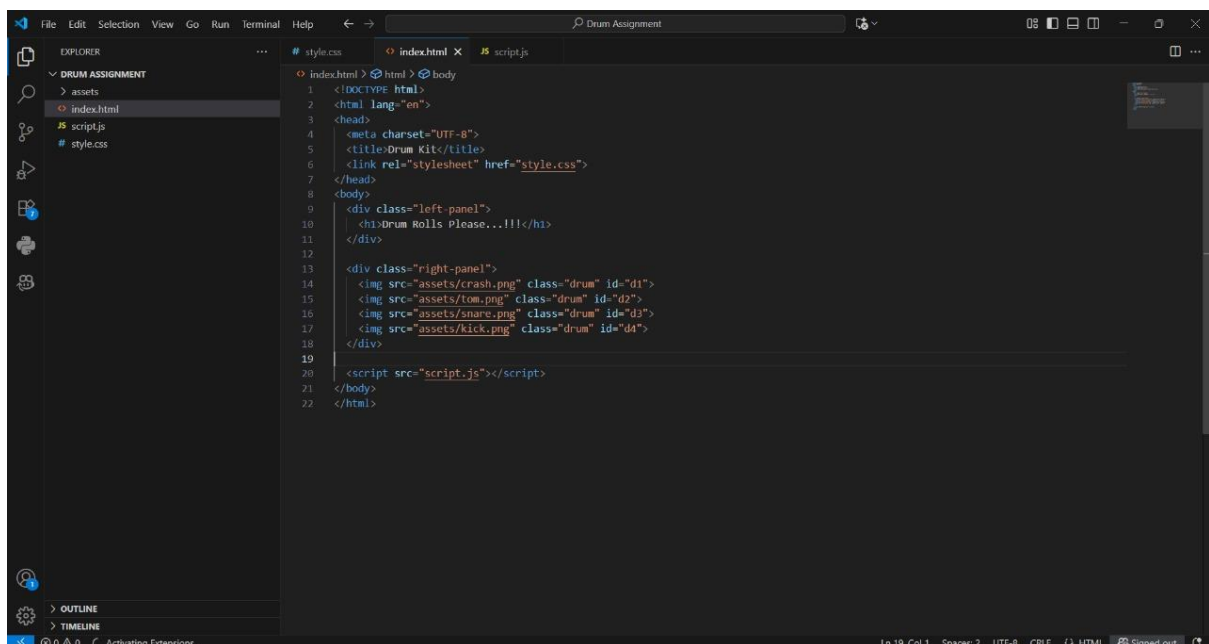
A web server is a computer or program that keeps website files like text, images, and videos and sends them to people when they visit a site. It uses rules called HTTP or HTTPS to share this information.. For example, when you open YouTube, its web server sends the videos and pages to your browser.

Q6.

Roles in web project:

- Frontend Developer: Creates and designs what users see on the website, like buttons, menus, and layouts, using HTML, CSS, and JavaScript.
- Backend Developer: Works on how the website functions behind the scenes, managing data, servers, and APIs using tools like Node.js, Python, or Java.
- Database Administrator: Takes care of storing, organizing, and protecting the website's data.

Q7.



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <title>Drum Kit</title>
6 <link rel="stylesheet" href="style.css">
7 </head>
8 <body>
9 <div class="left-panel">
10 <h1>Drum Rolls Please...!!!</h1>
11 </div>
12
13 <div class="right-panel">
14 
15 
16 
17 
18 </div>
19
20 <script src="script.js"></script>
21 </body>
22 </html>
```

Q8.

Difference Between Static and Dynamic Websites

Static websites display the same content to every visitor and are created using only HTML and CSS (e.g., a portfolio site).

Dynamic websites change content based on user input or data stored in a database (e.g., Facebook, Amazon). They use languages like PHP, Node.js, or Python to generate pages dynamically.

Q9.

Five Web Browsers and Their Rendering Engines

- Google Chrome - Uses Blink engine
- Microsoft Edge - Uses Blink
- Opera - Uses Blink engine
- Samsung Internet - Uses Blink engine
- Mozilla Firefox - Uses Gecko engine

Q10.

+-----+	+-----+	+-----+
Presentation Tier	Application Tier	Data Tier
(Web Browser)	(Server-side)	(Database)
+-----+	+-----+	+-----+
HTML, CSS, JS	Node.js, Python,	MySQL, MongoDB
UI / UX	Java	Data Storage
Client-side scripting	Business Logic	Data Access

When a user submits a form, the client sends a request to the server. The server processes it using business logic, interacts with the database to fetch or update data, and sends a response back to the browser, which displays the result. APIs act as bridges for communication between systems.

